

All the stakeholders and the other necessary people involved review the DDS by taking various factors into consideration, like the risks involved, the design modularity, feasibility, robustness, the cost of development, budget availability, time constraints etc. After rigorous brain storming, they choose the best design and that approach is selected for the product. Approval and sign-off is required by developer, customer, business analyst and tester (all stakeholders including inheritors). Focus shifts to the next process step only after approval.

Implementation / Coding

This is the implementation phase of the Software Development Life Cycle (SDLC). This phase is where the actual development or coding of the software is done. The development team starts its work by reviewing and understanding the design and SRS document. This is the most important phase for the software developers and the longest phase of the software development life cycle.

In this stage of the Software Development Life Cycle the product is developed. The code is programming as per DDS during this stage. If the design is clearly laid out in a crisp, clean, detailed and organized manner, the code generation can be accomplished without much hassle.

In case there are ambiguities in the software requirements, or if they are unclear or if the software development team is unable to understand user requirements correctly and further clarification is required, the user must be contacted and queried. Additionally, the software development team also returns the requirements that are understood by them.

Developers must follow the coding standards and guidelines defined by their organization. Numerous technologies are available for software development, each of which specialise in different scenarios. Therefore choosing the best one is important. Based on the requirements and the system design, appropriate technologies are chosen.

There might be cases when there is a need to bring a change in the hardware or software specifications. However, these changes are implemented only after the approval of the user. When the software code is completely written, it is compiled along with the other required files. Code inspection and reviews are conducted after the compilation. These methods are used to correct and verify errors in the software code. Software testing is carried out to detect and correct errors in each module of the software code. After the software code is tested, the software is delivered to the user along with the relevant code files, header files, and documentation files.